Collaborative and Configurable Thematic Itineraries

Project Description, Functional Specifications, Product Backlog and Detailed User Stories

Teaching Material for the Software Project Management Course

1. Context and Rationale

Enhancing a region's tourism potential requires integrating its historical, artistic, religious, cultural and natural heritage with experiences offered by local businesses. Traditional itineraries, which are often static, do not sufficiently encourage either business participation or visitor-led personalization.

The project envisages a collaborative platform: an editor defines a thematic route with fixed stops; businesses propose experiences located between two consecutive stops; and tourists select only the experiences they want, creating their own configuration of the route.

2. System Objectives and Scope

Students will design and develop a back office for administrators, editors and business operators, alongside a public-facing interface for tourists. The system must cover itinerary creation, registration of business operators, communication of opportunities, submission and evaluation of proposals, publication and final configuration.

- Editorial design: theme, description, geographical area, travel mode, estimated duration and sequence of fixed stops.
- Collaboration: identification of relevant operators, notifications and experience proposals associated with a specific segment.
- Personalization: selection of optional experiences without changing the fixed stops.
- The first version excludes payments, bookings and real-time availability management.

3. Core Domain Concepts

Thematic Itinerary and Fixed Stops

A thematic itinerary is an ordered route built around a defining theme, such as medieval abbeys, fortified villages, rural landscapes or Renaissance art. Fixed stops represent places of interest with a title, description and geographical coordinates. The editor defines them and sets their order; tourists may remove them but cannot rearrange them.

Segments and Experiences

A segment connects two consecutive fixed stops. For stops A, B, C and D, the segments are A–B, B–C and C–D. Each experience proposal must refer to one segment of a specific itinerary. Multiple operators may propose experiences for the same segment.

A business operator is a registered and active entity (restaurant, winery, workshop, accommodation provider, farm and similar businesses). An experience describes a service or activity, which may be paid; the proposal is the actual association of that experience with a segment of the itinerary.

Personalized Itinerary

This is a configuration derived from a published itinerary that includes only the stops and optional experiences selected by the tourist. The same thematic itinerary can produce many different configurations.

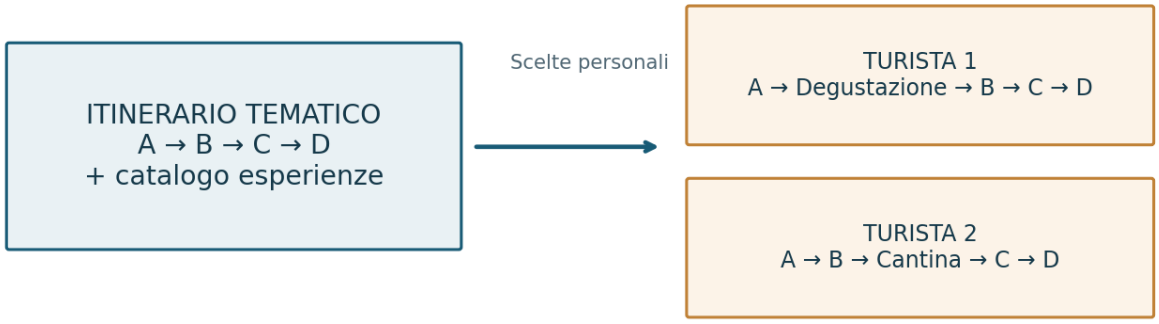


Diagram 1 — From the thematic route to different tourist configurations.

4. Collaborative Process

1. Creation

The editor creates the itinerary, enters its descriptive information, geolocates the fixed stops and determines their sequence.

2. Opening to Business Operators

The system identifies registered and active businesses near the route and informs them of the opportunity to participate.

3. Submission of Proposals

The operator selects a specific segment and proposes one or more experiences, providing a description, duration, price and terms.

4. Review and Publication

Proposals are reviewed according to the applicable rules; approved proposals enrich the published route.

5. Tourist Configuration

The tourist views the itinerary, selects the experiences they are interested in and receives a summary of their route.

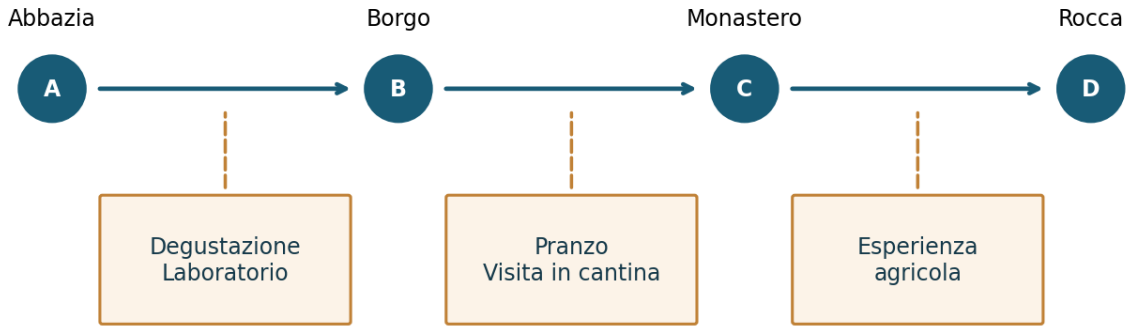


Diagram 2 — The five stages of the collaborative process.

5. Example Application

Itinerary "Abbeys and Medieval Villages": A — Santa Maria Abbey; B — San Pietro Village; C — San Benedetto Monastery; D — Medieval Fortress. The names are illustrative.

Segment	Proposed experiences
A–B	Tasting of local products; craft workshop
B–C	Traditional lunch; guided winery tour
C–D	Experience at a farm



Le esperienze opzionali sono associate ai segmenti tra tappe consecutive.

Diagram 3 — Example of fixed stops and optional experiences associated with segments.

One tourist may select a tasting and lunch; another may select a workshop and a winery tour.

6. Actors and Responsibilities

Actor	Responsibilities
Administrator	Users, roles, registration and activation of business operators.
Editor	Design of itineraries, stops and segments; evaluation of proposals and publication.
Business operator	Profile management and submission of experiences for specific segments.
Tourist	Browsing and personalization of published itineraries.

7. Back-Office and Tourist-Interface Functions

- Back office: authentication, permissions, master data, operator geolocation, creation of itineraries and stops, segments, notifications, proposals, approvals and publication statuses.
- Tourist interface: catalogue of published itineraries, stop details, experiences grouped by segment, selection, estimated total and personalized summary.
- Key constraint: an experience is proposed by an operator for a segment of a specific itinerary, not generically for the entire route.